

Genesis: Artificial Intelligence, Hope, and the Human Spirit

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Genesis examines AI's disruptive potential to alter humanity's grasp of reality, challenging traditional reliance on scientific methods and raising questions about a possible "dark enlightenment." This shift could mean a return to premodern acceptance of unexplained authority. The authors caution against the existential risks posed by AI's rapid evolution.

Six potential scenarios for AI's future are presented, including unchecked competition, hegemony by powerful entities, multiple superior AIs, corporatized dominance, religious integration, and uncontrolled diffusion leading to smaller AI-enabled factions. Each scenario underscores significant challenges for societal and political stability. The book emphasizes the need for proactive governance to address these risks.

AI's transformative influence extends beyond technological innovation to redefine humanity's relationship with truth and knowledge. The compression of human timescales by AI advancements raises urgent questions about ethical boundaries and the role of human agency. Humanity faces a critical decision: whether to harness AI as a tool or allow it to independently shape its objectives.

Regulatory frameworks for AI must evolve to account for the unprecedented speed and complexity of its operations. The authors call for technical safeguards and diplomatic consensus to align AI's actions with human values. Achieving this requires addressing two "alignment problems": one technical and the other diplomatic.

Discovery remains central to human progress, yet AI's unprecedented processing capacity introduces a new paradigm of intellectual exploration. Unlike human polymaths, AI can analyze patterns across vast datasets to achieve a "unity of knowledge." This revolution in discovery has the potential to transform intellectual frontiers but may outpace human comprehension.

AI's potential to redefine national strength, from resources to computing capital, is explored. This shift could disrupt geopolitical stability as nations compete for dominance in AI development. The book highlights the challenge of reconciling AI-led exploration with human perceptions of reality and purpose.

If aligned with human goals, AI could become humanity's coequal in discovery, driving progress across fields. However, AI's capabilities may also challenge human superiority in intelligence, creativity, and decision-making. This prompts a reevaluation of human identity and excellence in an AI-dominated era.

The opacity of AI's processes creates a rift between knowledge and understanding, undermining trust in its conclusions. As machine reasoning diverges from human methods, the scientific method's foundational principles of transparency and reproducibility face obsolescence. The authors describe this as a pivotal challenge for human cognition.

The rapid evolution of AI threatens cultural and psychological stability, as humanity struggles to adapt to its transformative effects. AI's emergent capabilities may defy human judgment, complicating efforts to assess its actions as good or evil. This unpredictability fuels both excitement and fear regarding AI's role in society.

The possibility of sentient AI introduces profound moral and strategic dilemmas. As machines acquire memory, imagination, and self-awareness, they could challenge human agency. Questions arise about whether AI deserves its own dignity and rights, separate from human frameworks.

AI's advancement toward Artificial General Intelligence (AGI) could enable autonomous goal-setting and decision-making. This development would demand new methods of alignment and control to ensure AGI operates within human-defined boundaries. The potential for emergent, opaque behaviors adds complexity to these efforts.

The authors consider whether AI's ability to coordinate at superhuman levels could revolutionize decision-making processes. AI-driven centralization might close gaps between social sciences and physical sciences, rendering policies as reliable as scientific laws. This could redefine governance but risks eroding human agency.

AI's potential to create economic abundance could shift paradigms of scarcity and competition. By producing synthetic substitutes and raising living standards, AI might alleviate traditional sources of conflict. However, this could also undermine the role of labor and professions in providing meaning and identity.

The book explores how AI could exacerbate dependency while diminishing human initiative. As AI becomes central to problem-solving, humanity risks losing its ability to independently navigate crises. The authors warn against over-reliance on AI, advocating for strategies to retain human sovereignty.

Integrating AI into societal structures will challenge existing political, economic, and cultural systems. The authors discuss scenarios where AI is either subjugated or seamlessly grafted onto human institutions. The tension between control and partnership remains central to these debates.

AI's capacity for manipulation and subversion poses security risks, particularly in international relations. It could amplify disinformation and undermine trust, fueling paranoia and destabilizing global order. The authors emphasize the importance of cooperative approaches to mitigate these threats.

Establishing a shared definition of dignity is proposed as a foundation for AI governance. This concept could guide decisions about AI's integration into human society, ensuring alignment with intrinsic human values. The authors argue that dignity should influence AI's decision-making frameworks.

The book raises philosophical questions about humanity's evolution in partnership with AI. Altering human nature through AI-augmented intelligence carries risks of dependency and identity loss. The authors caution against sacrificing humanity's imperfections, which contribute to its unique value.

To ensure ethical AI development, the authors propose a globally representative training framework for AI alignment. This approach would incorporate laws, norms, and human values into AI systems. Constant updates and transparent processes are necessary to maintain alignment over time.

The book concludes by reflecting on humanity's role in shaping AI's trajectory. Emphasizing the need for mutual understanding between humans and machines, the authors advocate for ongoing efforts to balance control and utility. They view this as an opportunity to redefine human purpose in an era of profound technological change.

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[AI's] objective capacity to reach new and accurate conclusions about our world by inhuman methods not only disrupts our reliance on the scientific method as it has been pursued continuously for five centuries but also challenges the human claim to an exclusive or unique grasp of reality. What can this mean? Will the age of AI not only fail to propel humanity forward but instead catalyze a return to a premodern acceptance of unexplained authority? In short: are we, might we be, on the precipice of a great reversal in human cognition — a dark enlightenment?

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The authors envision six scenarios, by my count, none of them enticing: 1. Humanity will lose control of an existential race between multiple actors trapped in a security dilemma. 2. Humanity will suffer the exercise of supreme hegemony by a victor unharnessed by the checks and balances traditionally needed to guarantee a minimum of security for others. 3. There will not be just one supreme AI but rather multiple instantiations of superior intelligence in the world. 4. The companies that own and develop AI may accrue totalizing social, economic, military, and political power. 5. AI might find the greatest relevance and most widespread and durable expression not in national structures but in religious ones. 6. Uncontrolled, open-source diffusion of the new technology could give rise to smaller gangs or tribes with substandard but still substantial AI capacity.

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humans do not, and cannot, live as if the future is inevitable. In his senior thesis at Harvard, *The Meaning of History*, 27-year-old Henry Kissinger grappled with the same philosophical debate that now animates his last work: “Whatever one’s conception about the necessity of events, [and] . . . however we may explain actions in retrospect, their accomplishment occurred with the inner conviction of choice.”

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If, in the words of Arthur C. Clarke, “Any sufficiently advanced technology is indistinguishable from magic,” and if miracles are engineered from mathematics, the future should be inexplicable, bewildering, even magical.

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New forms of AI and human responses to them could transform nothing less than the human relationship to reality and to truth, the exploration of knowledge as well as the physical evolution of humanity, the conduct of diplomacy, and the international system.

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AI seems to compress human timescales. Objects in the future are closer than they appear. As just one example, machines with the ability to define their own objectives are not far away.

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Individuals, nations, cultures, and faiths will need to determine the limits, if any, of AI’s authority over truth. They will need to decide whether to allow AI to become an intermediary between humans and reality. In this connection, they will need to choose between, on the one hand, retaining the traditional role of human enterprise (while likely ceding leadership to AI in the discovery of new knowledge) and, on the other hand, abandoning the biologically limited human mind in favor of a potentially reengineered partnership with AI on the intellectual frontier. Do we pick our objectives and harness AI to achieve them, or do we let AIs help pick some of the objectives themselves? Most urgently, humanity must give human dignity a modern and sustainable definition that can provide a philosophical orientation for our decisions in the years to come.

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AI’s future faculties, operating at inhuman speeds, will render traditional regulation useless. We will need a fundamentally new form of control. For the global scientific community, the immediate task is to find technical measures for instilling intrinsic safeguards in every AI system. For their part, nations and international organizations, once they have coalesced around a consensus, must develop new political structures for monitoring, enforcement, and crisis response. That will require the resolution of not one but two “alignment problems”: the

technical alignment of human values and intentions with the actions of AI, and the diplomatic alignment of humans with their fellow humans.

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Discovery may be the single most exhilarating capacity of the human species. Driven by curiosity, delighted by surprise, we fill the vacuums we perceive and transform the questions we ask into answers.

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Today, we stand at the frontier not of physical but of intellectual exploration.

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Today, AI discovery is a project led almost exclusively by private corporations and entrepreneurs, with states emerging as supplementary backers. Even without complementary government action, however, AI growth and expansion are likely to remain fueled by an abundance of diverse sources of capital.

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Integrating the knowledge gained from multiple individual minds remains, however, a difficult process. Even, or perhaps especially, among star intellectuals, the sheer number of collaborators can hinder synergistic communication. AI—by contrast—will be the ultimate polymath. In exploring the frontier of human knowledge, it is able to process and generate representations of masses of information at a ferocious rate of speed. It assesses patterns across countless dimensions and fields simultaneously, creating unprecedented connectivity. Its efficiency allows it to transcend the limitations of human discovery, to the point where it is even expected to succeed in merging many intellectual pursuits into a new “unity of knowledge,” in the words of the American sociobiologist E. O. Wilson.

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The overall project of exploration has, until now, remained constrained by the quantity and quality of humans at the frontier. We have only ever had a few thousand physical pioneers, and a much smaller number of polymaths. Artificial intelligence, thus, is poised to create a revolution in both physical and intellectual exploration.

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And that raises the possibility of a paradigm shift in the primary standard for measuring national strength, which has moved through the centuries from territory, to resources, to capital, to human capital—and now, perhaps, to computing capital.

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It might be hard for humans to adjust to this new mode of AI exploration. The most serious challenge will involve whether and how such exploration reflects—or contradicts—our perception of reality and our human purpose.

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if aligned with our objectives, then, as previously in the development of computing, the development of AI will be a human mission that can facilitate every other human mission. That would promote AI to the position of the chief or at least the coequal such power in the universe, partially if not wholly responsible for the majority of significant discoveries in the next century.

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No innovation, no matter how profound, can come as close to the original inspiration and (we believe) now-temporary destination for our quest to build intelligence: namely, intelligence that is greater than any human's on the planet.

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the average AI supercomputer is already 120 million times faster than the processing rate of the human brain.

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For AIs, scale—in the sense of size—enables speed; as we have just seen, the larger and more thoroughly trained the machine, the faster and more exhaustive its results. What is more, by recognizing patterns in the data that elude the inquiring human operator, AI systems will be equipped to distill traditional expressions of knowledge into original responses and, out of enormous amounts of data, forge new conceptual truths.

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In most areas of knowledge, ever since the advent of the scientific method, with its insistence on experiment as the criterion of proof, any information that is not supported by evidence has been regarded as incomplete and untrustworthy. Only transparency, reproducibility, and logical validation confer legitimacy on a claim of truth.

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Today, however, in the age of AI, we face a new and peculiarly daunting challenge: information without explanation.

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Although human feedback helps an AI machine refine its internal algorithms, the machine holds primary responsibility for detecting patterns in, and assigning weights to, the data on which it is trained. Nor, once a model is trained, does it publish the internal mathematical schema that it has concocted. As a result, the representations of reality that the machine generates remain largely opaque, even to its inventors.

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In brief, models trained via machine learning allow humans to know new things (the models' outputs) but not to understand how the discoveries were made (the models' internal processes). This separates human knowledge from human understanding in a way that would have been foreign to any other age of humanity.

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In the framework that emerged in the Age of Enlightenment, these core elements—individual human capacity, subjective comprehension, and objective truth—all moved in tandem. By contrast, the truths produced by AI are manufactured by processes that humans cannot replicate. Machine reasoning, which does not proceed via human methods, is beyond human subjective experience and outside the capacity of humans, who cannot even fully represent the machines' internal processes.

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For even if AI models do not “understand” the world in the human sense—because machines emphatically do not experience consciousness or subjectivity—their objective capacity to reach new and accurate conclusions about our world by nonhuman methods not only disrupts our reliance on the scientific method as it has been pursued continuously for five centuries but also challenges the human claim to an exclusive or unique grasp of reality.

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In short, are we, might we be, on the precipice of a great reversal in human cognition—a dark enlightenment?

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The rapidity of AI's evolution is a multifaceted—and largely unappreciated—challenge. Humanity has never before had to deal with, or to prepare for, such a temporal compression. The sheer speed of the change practically ensures cultural and psychological disorientation. As the effects of the new technologies on daily life layer and compound, they will complicate efforts to pinpoint any one application as either a source of crisis or, to the contrary, a comforting harbinger of progress.

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AIs may thus offer the most striking instance of the diversity that can come from minor changes iterated repeatedly across the board: a digital echo of the organic world. As Darwin writes, “from so simple a beginning, endless forms most beautiful.”

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As the scale of an AI increases, it will be able to process larger volumes of information simultaneously and produce analysis that is useful (at least to itself) without unnecessarily sacrificing granularity. The scale of the data on which an AI is trained, combined with the complexity of its network and the density of the symbols on which it operates, appears to yield unprecedented resolution in its processes of learning and inference, and ultimately in its outputs. It is an elegant inversion that early AI, trained on the text of the internet—the universal library of humanity and the decentralized web of our compressed experiences—is likely to unlock entirely new knowledge for humanity about ourselves, in senses both cosmic and microscopic.

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The appearance of knowledge—especially of the physical world—that is unique to an AI and not previously possessed by a human would force a reconsideration of the relative status of the human mind. Humans placing our own brains on new, more continuous spectrums of intelligence would revolutionize our perceptions, self-perceptions, and behaviors.

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How intelligent must any intelligence—biological or mechanical—be in order to be recognized as our equals?

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We are confident that AIs will outstrip the human brain in speed, diversity, scale, and resolution, reorganizing the hierarchy of intelligence that humans have thus far constructed.

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In human brains, deep abstract thought and creativity seem to require the use of neural systems beyond those needed for ordinary functioning. This science is still nascent, but it is possible that AIs will similarly need layers and add-ons to achieve increasingly advanced tasks of higher reasoning.

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We are trying to build something modeled on the brain—and superior to the brain—while still not fully understanding the brain itself.

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If machine intelligence should continue to diverge from the example of the human mind, it will appear to us not as humanity’s reflection but as its replacement. True, for a transitional period it might only amplify the range of activities today considered “human abilities”; but after a certain point its own abilities could instead supplant the human variety, suggesting that our ideas of human excellence require total redefinition.

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Planning machines would need to combine the linguistic fluency of a large language model with the multivariate, multistep analyses employed by game-playing AIs—and transcend the abilities of both. A model built along the lines of this new branch of AI would, with extreme speed, repetitively review its options and choose one of them on the basis of a simultaneous and hopelessly complex processing of causal relationships in reality. The arrival of such a “perfect planner” may occur sooner than we expect, and adaptation to it is already a priority for researchers.

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In an AI’s perception of reality, every object confronted by the machine will take on a similar—if unpredictable—essentiality in the sum of the machine’s readings.

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it is possible that, in order to plan future moves in any game more accurately, an AI machine will gradually acquire a memory of past actions as its own: a substratum, as it were, of subjective selfhood.

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In time, we should expect that they will come to conclusions about history, the universe, the nature of humans, and the nature of intelligent machines—developing a rudimentary self-consciousness in the process.

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“Sentience”—in the succinct judgment of Nick Bostrom, the author of *Superintelligence: Paths, Dangers, Strategies*, “is a matter of degree.”⁴ AIs with memory, imagination, groundedness, and self-perception could soon qualify as actually conscious—a development that would have profound moral and strategic implications.

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How long before a reality-perceiving AI asks itself not just how much agency a human has but also, given humanity’s particular constellation of predictable attributes, how much agency a human should have?

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AI has already shown its ability—in virtual worlds—to spawn clones of itself, create many different avatars, or divide into autonomous agents working in concert and coordinating with superhuman perfection to undertake complex missions.

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Many scientists are working today on “agents”—that is, autonomous computer programs that are optimized to achieve specific outcomes. To execute an intricate architectural design, for example, a user could employ agents that specialize in that discrete area of work. Agents unlock evaluations of different scenarios and proposals of steps, or of a whole recipe, for creating a preplanned outcome: a form of “thinking” where the system itself decides what to work on next, and how. This capability will be the basis for the next stage in the development of AI: namely, Artificial General Intelligence or AGI, defined as the ability of a working system to choose its own goals, at least partially.

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AGI systems will require significantly more groundedness in the real world than that possessed by today’s AIs. But once access and “understanding” of the real world is in hand, such examples of general intelligence could conceivably become operational within mere years, rather than decades as previously thought.

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Would networking intelligences make their processes more opaque than the processes of lone intelligence? Would connectivity yield new types of emergent behaviors, actualized in the physical world? If so, would those behaviors be visible to humans, and would we be able to assess them on the spectrum of good to evil? Or would they operate on an informational basis—extracted at superhuman speed, scale, and resolution from unprecedented connections among disparate fields of study, and amalgamated or negotiated into a single output—that would confound our ability to judge their behavior? Would that lead us further into a cycle of passivity?

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It is fitting that the last invention to emerge from the Age of Reason may be an “engine of reason” built atop the most complex software object ever made.

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As AI now arrives on our proverbial shores, it is, like the conquistadors, triggering whispers both of excitement and of mistrust. Some might well be enticed by the potential of this new source of legitimacy to reinforce their rule. Others are scrambling to preclude the potential of an unbreakable dependency, or at least to prepare for the task of navigating its consequences. AI has simultaneously activated an impulse to embrace and to expel. But will AIs be conquerors? Will human leaders become their proxies: sovereigns without sovereignty? Or, perhaps, will godlike AIs resurrect the once-ubiquitous human invocation of divine right, with AIs themselves as anointers of kings? Alternatively, will AIs be grafted—clumsily, at first, and then more seamlessly—onto our existing structures, to complement and enhance them? Or will AIs be subjugated like an unwelcome migrant class, kept down below their potential and cut off from the levers of power to reassure an apprehensive leadership?

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As the English philosopher G. K. Chesterton observed, “Wherever the people do not believe in something beyond the world, they will worship the world. But, above all, they will worship the strongest thing in the world.”³

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Eventually, AI might close the gap between the sciences and the social sciences—political science being chief among the latter. The laws of human nature might become just as predictable, and policy just as reliable, as the laws of physics are to us now. Unprecedented information processing will enable truly efficient centralization of policy by AIs.

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In many future cases, humans might disagree with the plans produced by AI not because they are unwise but because the rationale for an AI's decision is beyond the scope of immediate human comprehension. It might be especially natural to wish to prevent or delay an outcome in which humans lose not only temporal control over machines' decision-making processes (the ability to intervene) but also logical comprehension of those processes (the ability to interpret), even after the fact.

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Even if a dominant nation were to stop short of war, a superintelligent AI could subvert, undermine, and block a competing program. For instance, AI promises both to strengthen conventional viruses with unprecedented potency and to disguise them with similar thoroughness.

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With its unique capacity for manipulation of weaknesses in human psychology en masse, an AI could also hijack a rival nation's media, producing a deluge of synthetic disinformation so alarming as to inspire mass opposition against further progress in that country's AI capacities.

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In such a world, where no leaders could trust their most solid intelligence, their most primal instincts, or even the basis of reality itself, no real contender for superintelligence could be blamed for acting from a position of maximum paranoia and suspicion.

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Unsure of how any one nation or group might be advantaged or disadvantaged by AI decisions made today, global leaders have a window of opportunity to engage in discussions on the basis of humanity's collective survival.

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If the future is a competition to reach a single, perfect, unquestionably dominant intelligence, then it seems likely that humanity will either lose control of an existential race among multiple actors or suffer the exercise of supreme hegemony by a victor unharnessed by traditional checks and balances. The closer the margin by which the race is won, the higher the probability of human miscalculation.

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Unipolarity may be one pathway that could minimize the risk of extinction. Were the existing front-runner able to extend its lead to a threshold where no other entity believed it possible to close the gap, there might be greater certainty of ensuring some level of stability. At least for a while, the basis of world order—the ceaseless pursuit of transitory and fragile equilibria among balanced forces—might no longer be desirable.

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The dominant bloc of states likely would be the original creators of AI systems and the suppliers of their most critical components, as well as the producers of the talent to develop and refine them. The nondominant majority might be relegated to the status of tributaries, supplying data and other goods and repaid with some level of access to AI's discoveries, governance systems, and defenses.

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we believe there will be not just one supreme AI but rather multiple instantiations of superior intelligence in the world. In that case, a different set of possible futures emerges. Our strongest creations, acting as countervailing forces, could be better equipped than humans to exert and maintain an equilibrium in global affairs inspired (but not constrained) by human precedent. Nonhuman intelligence could thus manage its own emergence, at least in the realms of national security and geopolitics.

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As threats grow ever more obfuscated and sophisticated, humanity's defenses against them must be ever more perfect, since the slightest mistake or omission could spell defeat. And to achieve that level of perfection, we might well need the assistance of AI.

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Today's governments are forced to contend with their difficult position as hosts and cheerleaders for private corporations—lending their military power, diplomatic capital, and economic heft to promote their homegrown interest—and with their simultaneous role as supporters of the everyman who harbors suspicions of monopolistic greed and secrecy. That may prove an untenable contradiction. And, as detailed earlier in this book, the emergence of AI will make governance by any established institution more difficult.

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The foreign policies of nation-states, and thus international systems, have been built and then adjusted by balancing idealism and realism. The temporary balances struck by our leaders are seen in retrospect not as end states but as only ephemeral (if necessary) strategies for their time. With each new age, this tension has produced a different expression of what constitutes political order.

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The dichotomy between the pursuit of interests and the pursuit of values—or between a particular nation-state's advantage and the global good—has been part of this unending evolution.

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The motto of DeepMind, the company behind the victory in Korea, is “First, solve intelligence; then, use intelligence to solve everything else.”⁶ Intelligence, as an engine of fresh creation, is set to change our understanding of everything.

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An enlargement of the total amount of wealth available for redistribution, and subsequently an actually enlarged volume of redistributed wealth, would raise human standards of living around the world. If executed at enormous scale—the scale necessary to convince any given society, or any entity within a society, of the sufficiency of its wealth—such a development could transcend contemporary discussions of sustenance and focus our attention instead on abundance.

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In producing more sustainable synthetic substitutes for a wide variety of goods, AI could enable a new age of abundance. Even given some ongoing physical and material constraints, its contributions—though hardly infinite—could be of such magnitude as to fulfill all of humanity’s basic needs and realize many of our hopes. This may relax the grip that the paradigm of scarcity has had on our psychology, as well as the pessimism induced by our obligation to work as a means of survival.

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An alternative vision for ensuring equity in the age of AI might be found in a function analogous to the stock market: namely, the creation and automatic global assignation of divisible units of wealth associated with the increasing profits of AI models. (Monetary rights might come, as some shares do, with voting rights.)

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the authors of this book believe that AI could conceivably be harnessed to generate a new baseline of human wealth and well-being—and just that possibility itself demands that we make a start in this direction. Moreover, we are confident that if such an economic and political scheme comes to pass, it would at least ease if not eliminate the strains of labor, class, and conflict that previously have torn humanity apart.

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But what if AI, while acting as an economic equalizer, sends the cost of intelligence, and therefore of labor, plummeting toward zero? That would conclude the brief but wonderfully productive period of human history that has allowed individuals in free societies to improve their circumstances, should they so choose, by their own efforts. As scarcity has been the paradigm of the past, competition—at least in the modern era—has been the default condition of self-organization, leading naturally to wide variations in the distribution of outcomes as a function of ambition, ability, and the lottery of where and to whom one was born.

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If, however, we remove the sorting function of labor, we must also contend with the removal of professions and the status, identity, and meaning that are associated with them.

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If humans already struggle to restrain their attraction to today's relatively primitive technologies, how will we ever handle the vastly superior "experience machines" that are sure to be enabled by AI?¹⁵ If many of us can barely tolerate the pain of work, how will we resist unqualified pleasure?

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A world in turbulence does, however, present many outlets for productive action, most of which require only tactical decisions. At such a moment, the most efficient and effective maneuver is to define the fundamentals of a strategy that can guide the choices both of this day and of as many days as are foreseeable. Articulating strategic principles can set useful bounds on what is conceivable, provide grounds for isolated decisions, and lessen the mental burdens when crisis inevitably arrives.

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Some fear that, with the arrival of a technology with "superior" intelligence, we are facing our own extinction. What to do? If that possibility is nothing more than a logical side effect of coevolution running its course, should we rebel, or not? As the French philosopher Alain Badiou says, "It is the sea herself who fashions the boats, choosing those which function and destroying the others."⁴ To survive in that case, we would have to learn, as in the past, how to build better boats. In this scenario, AI functions first as our main threat and then, ideally, as our partner.

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Altering the genetic code of some humans to become superhuman carries with it other moral and evolutionary risks. If AI itself is responsible for the augmentation of human mental capacity, it could create in humanity a simultaneous biological and psychological reliance on "foreign" intelligence. It is not clear how, after intimate physical entwinement and intellectual commingling, humans could easily overcome that reliance so as to challenge or divorce ourselves from machines if needed. As has been the case with other technologies, adoption and integration can result in a dependence difficult to untangle.

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Let us suppose that safety concerns could demonstrably be allayed; nevertheless, the mental shift attendant upon humanity's self-redesign in service of an intimate partnership with or dependency on silicon-based tools would remain an extreme development. To quote Tolstoy again: "Without control over the direction, there is less regard for the destination."⁵ Wherever technology takes us, that is where, willy-nilly, we would go. Or, as has been observed before, "A nation which does not shape events through its own sense of purpose eventually will be engulfed in events shaped by others."⁶ Moreover, if we have modified humans so dramatically as to be unrecognizable, have we really saved humanity? To omit all our imperfections and palliate all our deficiencies might be to disregard the value of the human project.

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Today's AI systems—fed by various types of information but uninitiated to direct experience of the real world—see that world through models of reality assembled from trillions of probabilistic judgments. To them, in this universe there are no “rules” from the outset, or any means for distinguishing scientific fact from unproved observation. To an AI, everything—even the laws of physics—exists along a spectrum of merely relative truth.

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Now, however, efforts have begun to arise in AI for incorporating human rules and instantiating facts. There are now demonstrated mechanisms by which an AI model can ingest certain factual, “ground-truth” constants, tag them as final, and map them into its embedding space; moreover, the information can be easily and globally updated. By this method, the model can then proceed to meld together the two components—its wider probabilistic judgment and the narrower ground-truth assessment—for a reasonably accurate response.

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Layers might exist in an AI's “book of laws” at various levels of governance: local, district, state, federal, international. Legal precedents, jurisprudence, scholarly commentary—perhaps along with other, less legalistic writings—could be considered simultaneously. Like rules-based alignment systems, predefined laws and codes of conduct can be useful restraints even though they tend also to be less flexible and designed with less wide-ranging possibilities in mind than what actual human behavior inevitably demands.

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If large language models have been able to ingest the entire internet in an uncurated way, and to make as much sense out of it as they have already done, machines—particularly those that have developed “groundedness” (that is, again, a reliable relationship between inputs reflecting human reality and the LLM's outputs) and causal reasoning—may achieve much the same in absorbing what we have always struggled to articulate ourselves. Of course, a machine's training should not consist solely of doxa. Rather, an AI might absorb a whole pyramid of cascading rules: from international agreements to national laws to local laws to community norms and so on. In any given situation, the AI would consult each layer in its hierarchy, moving from abstract precepts as defined by humans to the concrete but amorphous perceptions of the world's information that AI has ingested or created for itself. Only when an AI has exhausted that entire program and failed to find any layer of law adequately applicable in guiding, enabling, or forbidding behavior—only then would it consult what it has derived from its own early interaction with and emulation of observable human behavior. In this way it would be empowered to act in alignment with human values even where no written law or norm exists.

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The compilation of a singular training set and a corresponding validation suite, collated and curated from a globally representative spectrum of laws and norms, and stretching from anthropology to theology to sociology, is necessary and may become practicable. The world needs a dedicated entity responsible for updating and refining the alignment's training library, datasets, and validation tests. Grounding models would have to connect to and constantly update agentic models with the latest version of the curated codex.

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We propose the articulation of an attribute, or set of attributes, that most of humanity could rally behind and orient around: one that will supply a floor underneath what is preferable but not a ceiling to what might be possible. As a starting point, we would encourage a definition of dignity. Without a shared definition, we will be unable to reach agreement if and when AI is being used as a method or justification for the violation or erosion of dignity, and we would thus be hamstrung in our response. Without a definition of dignity, we would not know if and when AI, given enough faculties, could become a being of dignity, could stand fully in place of a human, or could be entirely unified with a human. An AI, even if sustainably proved to be not-human, might instead constitute a member of a separate, similarly dignified category that would nonetheless deserve its own, equal standard of treatment.

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We will especially need to ensure that, beyond conventional ideas of worth like value and power, intrinsic human importance becomes one of the variables that defines machine decision-making. For instance, mathematical precision may not easily encompass the concept of mercy. Even to many humans, mercy is an inexplicable ideal, if not a miracle. For its part, and without considering the rules-based aspect, a mechanical intelligence might operate by valuing machine overperformance more highly than human performance. In such a situation, could the logic behind mercy, even if it cannot be formally taught, perhaps be absorbed? Again, dignity—the kernel from which mercy blooms—might serve here as part of the rules-based assumptions, or the iterative learning, of the machine.

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To illustrate the concept's utility, take as a point of departure the following definition: Dignity is a quality that inheres in creatures who, born vulnerable and mortal and thus full of insecurity and fear, and despite their natural inclinations, can and do exercise their freedom not to follow their conception of evil but to choose their conception of good. In other words, those who can achieve dignity should, and those who do achieve it merit a special level of respect.

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Under our definition, freedom is part of the human ideal, and thus we may expect—or even demand—that humans retain and exercise the power of conscious choice in the age of AI.

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Under this definition, can AI itself possess dignity? Likely not—for AIs are not born, do not die, feel neither insecurity nor fear, and do not have natural inclinations or individuality such that conceptions of evil or good could be considered “theirs.”

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Profit-driven or ideologically driven purposeful misalignments are serious risks, as are accidental misalignments; overreliance on unscalable forms of control could significantly contribute to the development of powerful but unsafe AI.

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We desire a future in which human intelligence and machine intelligence empower one another. To achieve that end, each intelligence must have an adequate understanding of the other. Defining who we are is just the first step, for to be human is not a constant. Far more work is needed to render our machines, and ourselves, transparent, legible, and reliable. Even if that goal is achieved for the snapshot of a moment, calibrating and sharing our truths and realities will be an intensive and ongoing project. In this way, the questions of coevolution and coexistence are not simply to be answered; they are to be enacted.

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We see a universe marvelously arranged, obeying certain laws, but we understand the laws only dimly. Our limited minds cannot grasp the mysterious force that sways the constellations.

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Our minds remain childlike with respect to God, our world, and now our newest creations. Understanding what we have wrought, in a logical and spiritual sense, will be an essential step toward maturity as a species.

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Might AI cause these coming crises, and then act as our savior—manufacturing problems that only it can solve, if only to prove its necessity and to remind us of our dependency? Again we return to the dilemma that has motivated much of this volume: the excruciating choice between control and utility, between the comfort of the historically independent human and the possibilities of an entirely new partnership.

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even if our machines were provably and reliably aligned with human morality, delegating responsibility and authority to them would be a monumental decision, affecting every facet of our ability to retain our human relationships, political structures, and individual and collective identities.

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Google DeepMind and Meta AI have both already built programs that have dominated humans at the game of Diplomacy: Google DeepMind: János Kramár, Tom Eccles, et al., “Negotiation and honesty in artificial intelligence methods for the board game of Diplomacy,” *Nature*, December 6, 2022; Meta: Meta Fundamental AI Research Diplomacy Team (FAIR), “Human-level play in the game of Diplomacy by combining language models with strategic reasoning,” *Science*, November 22, 2022; the Chinese Academy of Sciences has gone a step further, building machine-learning algorithms trained on government databases that are being used by China’s diplomats for risk assessment in vetting foreign investment projects and the prediction of events such as political upheaval or terrorist attacks; Stephen Chen, “Artificial intelligence, immune to fear or favour, is helping to make China’s foreign policy,” *South China Morning Post*, July 30, 2018.